DATA ITEM DESCRIPTION

Title: CHEMICAL AND PHYSICAL PROPERTIES FOR FORGINGS ANALYSIS REPORT

Number: DI-FORG-81196A
AMSC Number: 10588
DTIC Applicable: No
Preparing Activity: MR
Applicable Forms: N/A

Approval Date: 20250721
Limitation: N/A
GIDEP Applicable: No
Project Number: FORG-2025-005

Use/relationship: The Chemical and Physical Properties for Forgings Analysis Report provides information on the chemical composition and mechanical properties of forgings.

a. This DID satisfies the requirements of paragraph 3.13 of MIL-DTL-46172.

b. This DID supersedes DI-FORG-81196.

Requirements:

1. Reference Documents. The applicable issue of the documents cited herein, including their approval dates and dates of any applicable amendments, notices, and revisions, shall be as specified in the contract.

2. Format. The Chemical and Physical Properties for Forgings Analysis Report format shall be contractor selected. Unless a specific format is essential for compatibility, the initially used format arrangement by the contractor shall be used for all subsequent submissions.

3. Content. The Chemical and Physical Properties for Forgings Analysis Report content shall contain the data elements listed within the included figure 1 sample as follows. The figure 1 sample format is not mandatory, and contractors may use their own preferred data listing formats provided all required data elements are present.

3.1 General Information. The report shall include the RECORD OF TREATMENT AND TEST OF (enter appropriate identification), Contract (name and address of contractor), Contract order No., Date, Specification No., Drawing number, Forging No., Heat No., Date Forged, Shipped To (Released), and Date Shipped, Released.

3.2 Chemical Analysis. The report shall list the complete Obtained and Required chemical composition (C - carbon, Mn - manganese, P - phosphorus, S - sulfur, Si - silicon, Ni - nickel, Cr - chromium, Mo - molybdenum, V - vanadium, blank - other) of the forging.

3.3 Physical Information. The report shall include Macroetch satisfactory, Macroetch not taken, Cold Worked and Soaked, Not Cold Worked, Ingot No., Type of Ingot, Ingot Size, Length (inches) Total Weight, Wt. (weight) Above Piece, Wt. (weight) Below Piece, Date Forged.

3.4 Heat Treatment Procedure. The report shall include a complete and detailed description of the heat treatment procedure including the date completed.

3.5 Mechanical Properties. The report shall include specimen locations, Y.S. (yield strength), T.S.

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(tensile strength), Elon Percent (percent elongation), R.A. (reduction of area), Hardness, Fracture, and Charpy (ft-lb) values.

3.6 Authentication. The Chemical and Physical Properties for Forgings Analysis Report must include identification of the person performing the physical tests (PHYSICAL TESTS MADE BY) and must be signed by the appropriate government inspector with the place and date identified.

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CHEMICAL AND PHYSICAL PROPERTIES FOR FORGINGS ANALYSIS REPORT

RECORD OF TREATMENT AND TEST OF:

Contract:	**Contract Order No:**	**Date:**
Specification No.	**Drawing No.**	
Forging No.	**Heat No.**	
Date Forged:	**Date Shipped, Released:**	

SHIPPED TO:		**PHYSICAL INFORMATION**
Name:		**Macroetch Satisfactory:**
		Macroetch not taken:
Address:		**Cold Worked and Soaked:**
		Not Cold Worked:
		Ingot No.
		Type of Ingot:
POC:		**Ingot Size:**
Phone No:		**Length (inches):**
Email:		**Total Weight:**
HEAT TREATMENT PROCEDURE		**Weight Above Piece:**
		Weight Below Piece:
		Date Forged:
		Remarks:
Date Completed:		

MECHANICAL PROPERTIES

Specimen Location	YS	TS	Elongation %	Area Reduction %	Hardness	Fracture Type [1/]

Charpy Location	1 (ft-lbs)	2 (ft-lbs)	3 (ft-lbs)	4 (ft-lbs)	5 (ft-lbs)	6 (ft-lbs)

Physical Tests Performed By:

CHEMISTRY ANALYSIS

	C	Mn	P	S	Si	Ni	Cr	Mo	V	
Obtained:										
Required:										

Chemical Analysis Performed By:

DO NOT FILL BELOW THIS LINE - TO BE COMPLETED BY CERTIFIED TEST FACILITY OFFICIAL

[1/] Fracture Type Abbreviations:	Location:	Date:	Certified Test Facility Inspector:
C - Cupped Cr - Crystalline I - Irregular P - Pitted L - Laminated N - Nonmetallic W - Woody Lc - Lustrous Cavities			

FIGURE 1. Sample Format.

End of DI-FORG-81196A

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